PART II — FROM PHYSICAL TO DIGITAL · CHAPTER 07

Interfaces as Tolerances

07

An interface takes shape within the limits of materials, processes, device, context, business and person. Designing means recognizing those limits, measuring them, and deciding who absorbs the effort when we exceed them.

Oct 19, 2026 · 20 min read · Living edition 0.7 · Status: in progress ·

When you work in industrial design, you soon discover that a user's interaction with a product begins before the object exists. It begins with knowing which materials you can source, which machine you will produce on, and which mold you can afford. Many of the sensations we later credit to design are decided there: how firm or soft a grip feels, where one part ends and another begins, what it costs to repeat the object ten thousand times.

A toothbrush shows that whole journey, because it is an everyday contact between hand, material and body. Almost nobody thinks about its mold, and yet every manufacturing decision reaches the mouth of the person using it. I use it as an example of something anyone can check against the object in their own bathroom.

The question I want to carry from there into the digital world is simple. First: what can we make? Then: how can someone use it? And finally, the one that weighs most once the product is in a person's hands: what are we demanding of them when they use it every

day? Chapter 5 introduced digital materials and Chapter 6 their preservation over the life of a product;[6] this one examines how those decisions reach the person through the interface.

PART II — FROM PHYSICAL TO DIGITAL · CHAPTER 07 · 01 MATERIAL AND PROCESS

What a Toothbrush Allows



Figure 7.1 — Rigid body, soft grip zone and bristles: three decisions the user feels as one.

Consider a toothbrush made of a rigid body and softer grip zones. That combination is not decided on the drawing alone. Multi-component injection, the mold and the compatibility between materials shape its geometry, its joints and its surface: where the soft part can end, what thickness it allows, how it bonds to the body and what texture remains visible. The machine bounds what is possible; the design team still makes decisions inside those bounds.

It helps to separate that body from what happens with the filaments. The documentation from ZAHORANSKY, a manufacturer of brush-making machinery, distinguishes the injection of one- or multi-component parts from the placement of the filaments and their finishing, which includes trimming and rounding the tips.[1] They are different processes and each one delivers different decisions: the first defines the grip and the shape; the others define what touches gum and enamel. Knowing which one a decision belongs to keeps us from blaming the mold for

what is the filament's, and the filament for what is the mold's.

The same goes for hard, medium and soft bristles. The commercial categories are useful for talking about perception, but the real response depends on a combination of material, diameter, length and arrangement of the filaments. Two brushes that both say "medium" can feel different. I am not comparing specific brushes here; I use the categories only to show that a sensation is the consequence of several variables at once.

To follow the argument it helps to fix three words with a simple meaning. They resemble one another, which is why they get confused.

TABLE 7.1 — THREE TERMS THAT ARE NOT SYNONYMS

TERM	WHAT IT MEANS IN THIS CHAPTER	IN THE TOOTHBRUSH
Constraint	A condition that bounds a decision, such as the available process, the space of a room or the budget	The mold you can afford and the materials that are compatible with each other
Technical tolerance	Permitted variation from a specification, for example in a dimension or in the fit between parts	The admissible variation in a wall's thickness or in the fit between the body and the soft zone
Acceptance threshold	The limit of an admissible response, such as actuation effort, the intensity of an alert or the duration of a task	How much firmness the user tolerates before it becomes bothersome

This chapter borrows the idea of tolerance from manufacturing and extends it to interaction design. The availability of a material is a constraint; its properties and permitted variations help establish its operating margin. People also show differences and limits that must be studied. That extension is an editorial proposal of this chapter: they are not technical synonyms and I do not use them as if they were.

Let us push the logic to the extreme. With the right machinery and materials it is possible to manufacture a toothbrush with metal bristles. Nothing in the process prevents it. And yet being able to make it does not show that the combination is appropriate for the intended contact: that a way to produce something exists does not answer whether it should exist. This is the point at which the possibility of production stops being enough to justify a design.

The user feels the consequence of these decisions every time they pick up the object and use it. In an application something similar happens, even if its materials are less visible.

Being able to make a toothbrush with metal bristles does not show that the combination is appropriate for the intended contact.

— Chapter 07 · 01

An interface takes form within what materials, software runtimes and

human biology permit. Good design does not pretend these constraints do not exist; it establishes precise margins within which the experience remains humane and reliable.

One Route, Three Interfaces



Figure 7.2 — Car: attention is shared with driving.



Figure 7.3 — Phone: one hand occupied and daylight on the screen.



Figure 7.4 — Tablet: unhurried browsing to plan a trip.

Take a need everyone recognizes: finding your bearings and getting somewhere. A navigation service such as Google Maps serves it in a car, on a phone and on a tablet, and the need is the same in all three cases. What changes is everything surrounding the person who has it. I do not promise that every current version of the service offers exactly the same functions on every device; I use the example to compare scenarios, not to describe a product.

In a car, attention is shared with driving. Google's design guidance for driving (Android Auto / Android Automotive OS) establishes that driving-related tasks take priority over any other, that information must be readable at a glance — it proposes a maximum of two seconds to read a piece of content and return to the road — and that no operation should require taking both hands off the wheel.[2] These are limits of context, not of screen: the same route calls for fewer steps, shorter instructions and voice.

On a phone, during a walk, the hand is occupied, daylight competes with the screen, battery matters and attention is divided with the surroundings: the street, other people, the next crossing. What the interface must achieve is that the next action is recognizable and can be handled with one hand.

The tablet, by contrast, tends to appear when someone is preparing a trip. There is more surface and room for unhurried browsing: comparing alternatives, seeing more context. But it is wise not to assume that every tablet has the same sensors or serves the same purpose as another.

TABLE 7.2 — THE SAME NEED IN THREE SITUATIONS

SCENARIO	PRIMARY CONSTRAINT	WHAT THE INTERFACE ADJUSTS	RISK IF IT IGNORES THE CONTEXT
Car	Divided attention and limited glance duration	Voice prompts, large targets, essential route guidance only	Driver distraction, delayed reactions, road danger
Phone on foot	Single-hand operation and direct sunlight	High contrast, immediate next-turn cards, haptic feedback	Stops on crowded sidewalks, disorientation, missed turns
Tablet at desk	Exploration and side-by-side comparison	Dense overview, alternate routes, rich venue metadata	Cluttered cognitive load if carried into mobile contexts

What People Can Absorb

When an interface demands more than the human senses can process, the person begins to make mistakes. This is not carelessness; it is biology. Visual acuity drops under poor lighting, auditory discrimination degrades in noisy environments, and working memory holds only a few chunks of volatile information at once.

Industrial design learned this lesson through physical switchboards and aircraft cockpits: grouping related controls, providing distinct tactile profiles, and establishing unambiguous confirmation signals prevented catastrophic operator error. Digital interfaces inherit the exact same human receiver.

In aesthetic and perceptual studies, the concept of MAYA ("Most Advanced, Yet Acceptable") studied by Paul Hekkert and colleagues demonstrates that human preference balances on an edge between novelty and typicality.[3] Push an interface too far into unfamiliar interaction metaphors, and users abandon it; keep it purely conventional, and it fails to leverage new technical capabilities. The tolerance band is narrow and measurable.

Consider audio feedback. Consumer products such as Marshall headphones rely on a distinctive guitar-riff cue to confirm Bluetooth pairing.[4] The sound is brief, instantly distinguishable from speech or notification chimes, and unmistakable in its meaning. It respects the ear's tolerance for interruption while providing definitive confirmation.



Figure 7.7 — Immediate confirmation: unambiguous feedback acknowledges task completion without cognitive hesitation.

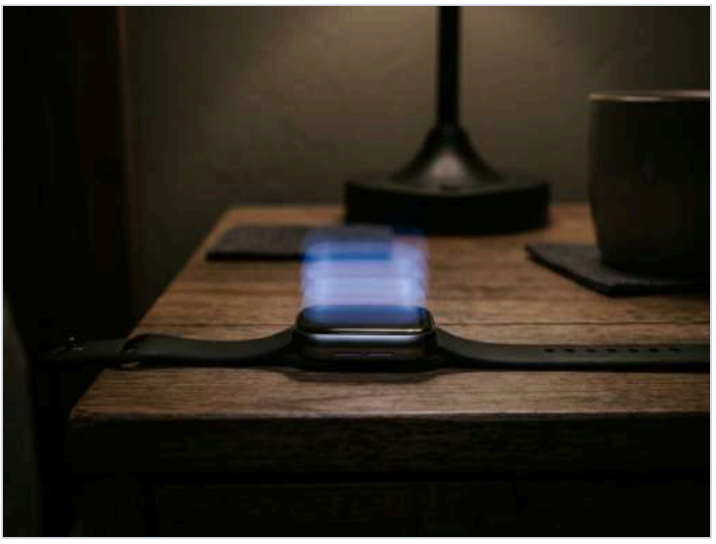


Figure 7.8 — Controlled repetition: consistent cues allow rapid scanning without re-interpreting syntax.





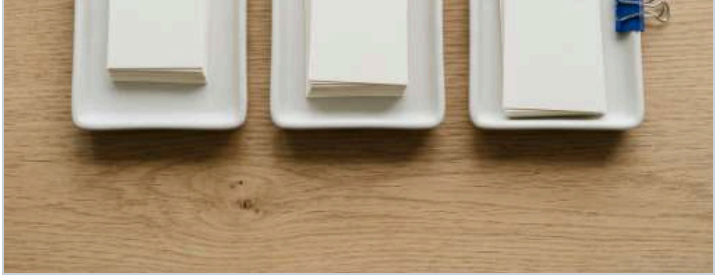


Figure 7.9 — Functional grouping: spatial proximity binds related controls, reducing eye saccades.



Figure 7.10 — Physical control differentiation: distinct geometries prevent accidental actuation under stress.

The Envelope of Movement

An interface does not float in a vacuum. It lives in physical rooms, vehicle cabins, warehouse aisles, or hospital wards. Every physical object claims an envelope of movement — the spatial volume required not just to place the object, but to enter it, adjust it, clean it, and safely exit.

Consider an aircraft passenger seat or high-density auditorium chair. The seat itself occupies a fixed rectangular footprint. But the operational envelope includes the recline angle, leg clearance, tray table deployment, and aisle egress.

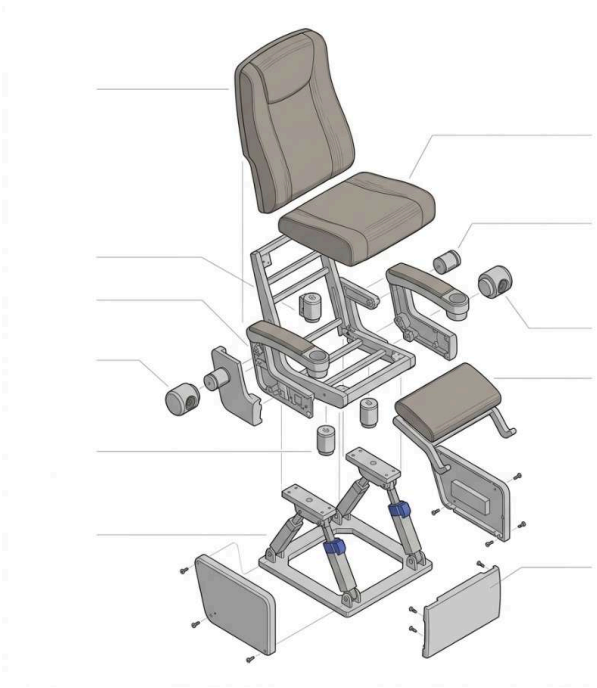


Figure 7.5 — Exploded view of a modular seat: the mechanical tolerances between chassis, pivot hinges and cushioning dictate the user's envelope of movement.

When interior planners configure a room, they face an unavoidable trade-off between nominal capacity and operational tolerance. Cramming maximum seats into a hall (Layout A) reduces aisle clearance to the bare legal minimum, creating bottlenecks during ingress and making cleaning tedious. Providing generous margins (Layout B) lowers nominal capacity but preserves rapid circulation and dignity of movement.

Figure 7.6 — Spatial layout

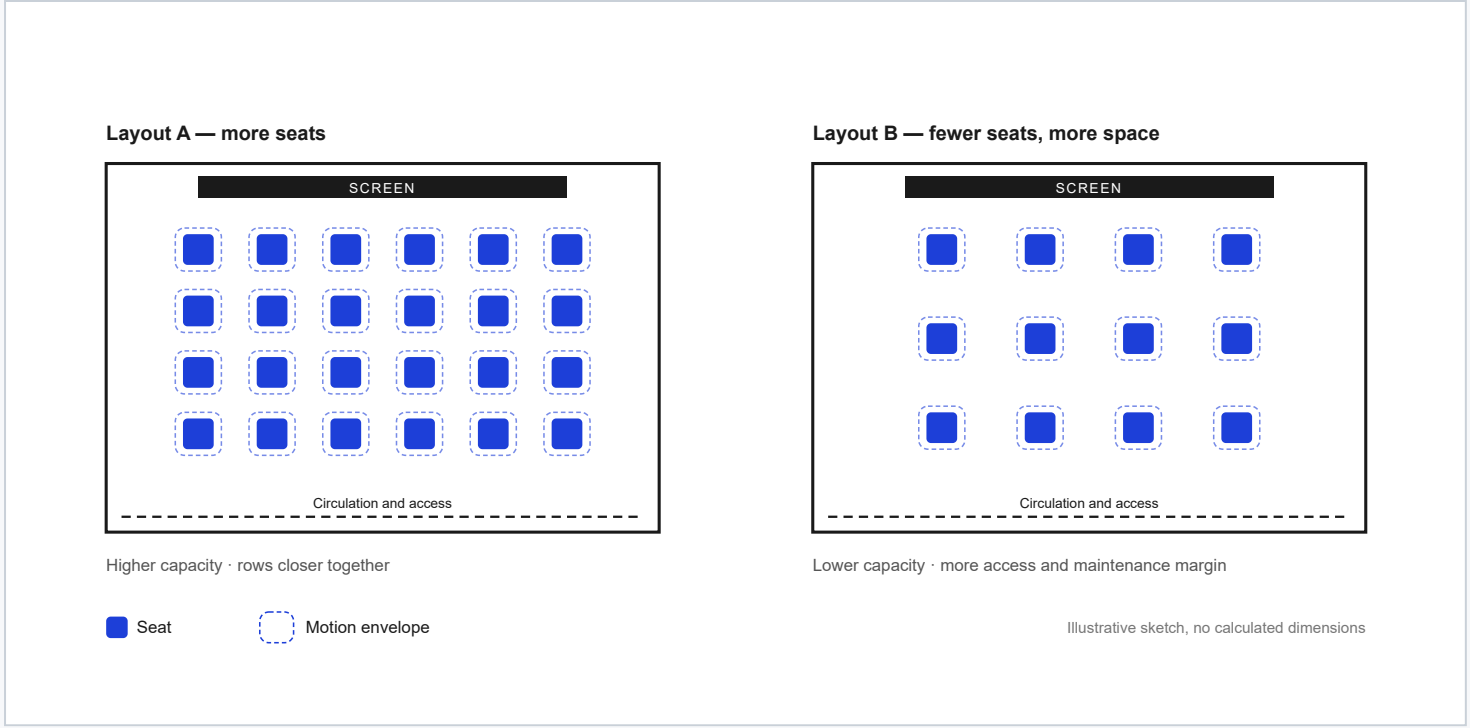


Figure 7.6 — Two room configurations: Layout A prioritizes nominal seat capacity at the expense of circulation; Layout B sacrifices row density to preserve maintenance clearance and swift egress.

The Cost of Cumulative Repetition



In logistics facilities, retail fulfillment hubs, and hospital pharmacies, warehouse workers and operators scan hundreds of packages every hour. A handheld scanner weighing 450 grams might seem trivial during a ten-minute executive demo. But repeat that scanning motion 2,000 times across an eight-hour shift, and that weight becomes an ergonomic hazard.

The Occupational Safety and Health Administration (OSHA) notes that awkward postures, repetitive motions, and forceful exertions are primary risk factors for

Figure 7.11 — Handheld barcode scanner: pistol grip vs. top-down grasp. The angle of actuation dictates cumulative strain across thousands of cycles.

musculoskeletal disorders in warehousing.[5] When an interface forces excessive wrist flexion or thumb hyperextension to reach an “Enter” touch-target, the product is directly offloading engineering shortcuts onto the biological joints of the worker.

Ergonomics is not an aesthetic preference; it is a structural tolerance. If our software flows require three confirmation taps instead of one automated trigger, we are multiplying mechanical joint cycles by thousands each working day.

When System Limits Collide

Product leaders rarely get the luxury of optimizing a single tolerance in isolation. Industrial engineering is the discipline of negotiating conflicting constraints: weight versus battery longevity, tactile feedback versus waterproof sealing, visual richness versus rendering latency.

In software interfaces, the tension between screen real estate and cognitive clarity follows the same law. Packing an enterprise dashboard with sixty live data widgets satisfies an executive’s desire for comprehensive visibility, but it overwhelms the operator’s visual field, increasing error rates during critical incidents.

Establishing deliberate tolerances means drawing boundary lines: what is the maximum number of actionable alerts allowed per hour? What is the maximum acceptable latency between physical barcode scan and server confirmation? When these thresholds are defined as hard contractual constraints rather than aspirational guidelines, product teams design defensively.

How Interfaces Break Down Under Stress

A bridge does not fail on a calm, sunny day under nominal traffic. It fails when heavy winds, thermal expansion, and peak resonant loads align. Similarly, digital interfaces do not fail during rehearsed sales demonstrations; they collapse when network latency spikes, sunlight washes out the display, and an exhausted user is rushing to meet a deadline.

When an interface exceeds its tolerance band, it enters one of three distinct failure modes: silent error, user adaptation fatigue, or catastrophic system rejection. In silent error, the user believes an action succeeded when it actually failed silently in the background. In adaptation fatigue, the human compensates for poor ergonomics until chronic strain forces sick leave. In system rejection, operators actively bypass the application using unauthorized shadow tools.

TABLE 7.3 — FOUR OPERATIONAL FAILURE MODES

FAILURE MODE	UNDERLYING TOLERANCE EXCEEDED	OBSERVABLE SYMPTOM	ROOT REMEDIATION
Silent Dropped Input	Touch sampling rate & network debounce threshold	Operator taps submit twice; duplicate orders or inventory discrepancies	Hardware haptic confirmation and deterministic client-side queuing
Alert Desensitization	Auditory & cognitive alarm threshold	High-priority alarms ignored alongside mundane system notifications	Strict hierarchical triage; cap non-actionable warnings
Physical Fatigue Bypass	Repetitive wrist torque & device mass threshold	Workers set scanner down on cart and enter 12-digit barcodes manually	Switch to wearable ring scanner or optical fixed-mount reading
Glance Latency Distraction	Visual glance duration threshold (> 2 seconds)	Drivers miss exits or swerve while attempting to navigate complex menu trees	Flatten visual architecture; enforce voice-driven execution

What Exceeding the Margin Costs the Business

Bad interface design is often dismissed as a cosmetic defect or a matter of user preference. In high-volume operations, however, fractional ergonomic friction translates into massive corporate liability.

Consider an enterprise distribution network with 250 warehouse picking operators. In a typical month, the team executes 600,000 package barcode scans (hypothetical model). If poor scanner weight balance and ambiguous on-screen confirmation generate a 0.02% error rate — equivalent to 120 misdirected packages per month — and each misrouted parcel costs an average of $42 USD to locate, restock and reship, that tiny interface imperfection extracts over $5,000 USD in monthly waste, or $60,000 USD annually.[7]

Add to that the unseen costs of worker turnover, musculoskeletal compensation claims, and productivity loss, and the true price of "tolerating bad tolerances" dwarfs the engineering investment required to fix it.

PART II — FROM PHYSICAL TO DIGITAL · CHAPTER 07 · 09 METHOD

Measure Before Asking People to Adapt

Identify, measure, decide, validate: a four-stage method that repeats every time the hardware, the flow or the pace of work changes.

TABLE 7.4 — SIX LAYERS OF THE MAP OF LIMITS

LAYER	EXAMPLE VARIABLE	EVIDENCE NEEDED TO DECIDE
Material and process	Variation in a part or grip response	Specification and manufacturing tests
Device and context	Actuation, reading and steps during the task	Tests on the intended device and in the intended situation
Sensory and cognitive	Alert frequency, errors and recovery	Research with diverse users and representative exposure
Space and operation	Spacing, movement and access	Validation of the installed whole

LAYER	EXAMPLE VARIABLE	EVIDENCE NEEDED TO DECIDE
Economics	Effective use, cost and incremental value	Scenarios and adoption tracking
Health	Postures, force, repetition and duration	Competent ergonomic evaluation and response to reported discomfort

01

Identify

Recognize the exposure before measuring: which gestures repeat, for how long, in what posture and at what pace.

02

Measure

Measure in representative conditions with diverse bodies and experience, looking at spread and not only average.

03

Decide

Decide the adaptation based on verifiable requirements; never pass the adaptation on to the worker.

04

Validate

Validate real sustained use week after week, evaluating again after changes in flow or pace.

What the User Ends Up Paying

That person's health must have an explicit place in the requirements, in the tests and in

our product decisions.

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Back to the toothbrush. The metal bristles made an inadequate decision visible: nobody needs a study to see that they should not touch a gum. In a digital or hybrid product the effort is not so easy to see, because it is spread across alerts, gestures, grips, waits and corrections, each small and all repeated.

The responsibility belongs to the product. It means bringing health and everyday exposure into device selection, into the definition of requirements and into the acceptance of the implementation. A successful demo needs to be complemented with evidence of representative use. And when tests reveal an avoidable load, what is called for is to revisit the object, the flow or the operation before passing on to the person the obligation to adapt.

As product managers, we have to look at what happens after the first interaction. That device will stay in someone's hand throughout their working day, with the same gestures and the same demands we defined when we implemented it. Chapter 8 follows that trail: it is devoted to what changes when a product scales.

REFERENCES & NOTES

1. **ZAHORANSKY** — Personal Care. Manufacturer documentation distinguishing the injection of one- or multi-component parts, the placement of filaments and their finishing (trimming and rounding of tips). ↩
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6. **This book** — Chapter 05 (Digital Materials: APIs, Data and Constraints) and Chapter 06 (Designing for Maintenance, Not Just Launch). ↩
7. **Editorial note** — The proposed structure, scenarios, conceptual seat, room layout and economic calculations are the author's elaboration. Figures and monetary values are hypothetical models for comparative analysis. ↩

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08 — What Breaks When You Scale a Product — in development

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